Milestone 2 - Model Integration & Baseline Results

MuseumCaption: Style-Conditioned Image Captioning using CLIP + GPT-2

Neural Network and Deep Learning - Project 2

Group 5

Aditya Kumar

Meghana Rao

Simran Abhay Sinha

Sushmitha Sudharsan

kumar.aditya1@northeastern.edu

rao.meg@northeastern.edu

sinha.sim@northeastern.edu

sudharsan.s@northeastern.edu

Signature of student 1: Aditya Kumar (25%)

Signature of student 2: Meghana Rao (25%)

Signature of student 3: Simran Abhay Sinha (25%)

Signature of student 4: Sushmitha Sudharsan (25%)

Github

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1. Training Log

The milestone 2 work focused on connecting the CLIP image encoder with the GPT-2 language decoder through prefix-based embedding injection. The model uses CLIP image features as input to a trainable projection layer, and the projected prefix embeddings are concatenated with GPT-2 token embeddings during caption generation.

Environment and setup:

- Runtime device: NVIDIA A100-SXM4-40GB GPU
- CLIP model loaded from cached milestone 1 files
- GPT-2 model loaded from cached milestone 1 files
- Dataset loaded from cached Flickr30k subset
- Available image-caption pairs used in this run: 1000
- Available image embeddings loaded from cache: 1000

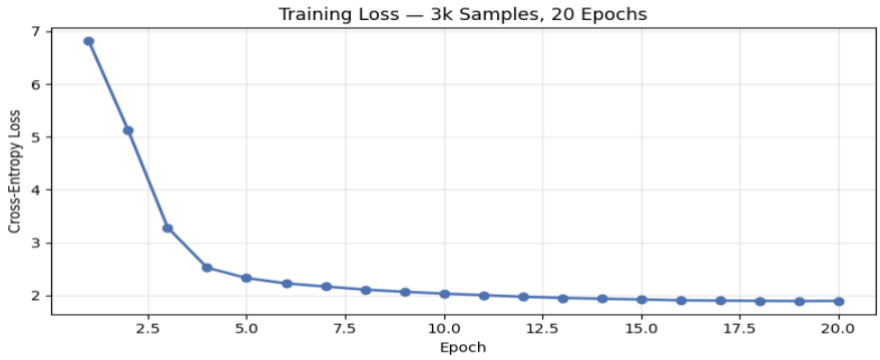
Model integration details:

- Image encoder: CLIP
- Language decoder: GPT-2
- Embedding injection method: Prefix conditioning
- Prefix projection parameters: 3,939,840
- GPT-2 parameters were frozen during training
- Only the prefix projection layer was trained

Training configuration:

- Batch size: 64
- Optimizer: AdamW
- Learning rate: 2e-4
- Learning rate scheduler: CosineAnnealingLR
- Maximum caption length: 32
- Number of planned epochs: 20

Training progress: The notebook resumed from a saved checkpoint at epoch 20. The saved loss curve shows that the training loss dropped across epochs and then leveled off near the end of training.



Generated caption setup:

- Prompt prefix used during generation: “a photo of”
- Maximum new tokens: 60
- Minimum new tokens: 15
- Beam width used in current code: 5
- no_repeat_ngram_size: 3
- length_penalty: 1.5

2. Early Caption Samples : The following examples show early caption outputs generated by the CLIP + GPT-2 model after training the prefix projection layer. For reference, the code also compares the outputs against BLIP and the original Flickr30k caption.

Sample 1

Original caption: The man with pierced ears is wearing glasses and an orange hat.
CLIP + GPT-2 caption: a man wearing a shirt with the words "I'm a man" in the middle of his face.

BLIP caption: a man wearing a hat and glasses with a cell in his hand.

Sample 2

Original caption: A black and white dog is running in a grassy garden surrounded by a white fence.
CLIP + GPT-2 caption: a dog with a dog collar on its back and a dog leash on its front.
BLIP caption: a dog running in a grassy field with a white picket fence in the background.

Sample 3

Original caption: A young female student performing a downward kick to break a board held by her Karate instructor.
CLIP + GPT-2 caption: a man with a helmet on his head, wearing a helmet and a pair of gloves on his feet.
*BLIP caption:*a red, blue, and white rug.

Sample 4

Original caption: Five snowmobile riders all wearing helmets and goggles line up in a snowy clearing in a forest in front of the trees.
CLIP + GPT-2 caption: a man and a woman wearing a ski mask and a backpack with a backpack on their shoulders and a ski hat on their heads.
BLIP caption: a group of people standing in the snow.

Sample 5

Original caption: Two men sitting on the roof of a house while another one stands on a ladder.
CLIP + GPT-2 caption: a man and a woman sitting on a bench in front of a fire hydrant.
BLIP caption: a blue sky with a few white clouds.

3. Summary of Observations

In this milestone, the CLIP image encoder was connected to the GPT-2 language decoder using prefix conditioning. The image embedding from CLIP was passed through a trainable projection layer, and the projected output was injected into GPT-2 as a prefix before text generation. GPT-2 itself was kept frozen, so the training focused only on learning the mapping from image features into the decoder input space.

The training log shows that the loss dropped a lot in the first few epochs and then moved more slowly after that. The saved curve starts near 6.8 in the first epoch and reaches about 1.9 to 2.0 by the end of training. This shows that the prefix projection layer was learning a usable connection between the CLIP embeddings and GPT-2 token space.

The early caption samples show that the model can produce sentence-like outputs. In several examples, the model repeats object types, mixes up people and actions, or adds details that are not present in the image. This can be seen in outputs such as repeated helmet references, wrong scene descriptions.

The BLIP baseline performs better than the CLIP + GPT-2 model on most of the sample images in this run. BLIP usually stays closer to the image content, even when the full scene is not captured. The CLIP + GPT-2 model, on the other hand, shows that the integration works at a system level, since it produces valid text from image-conditioned embeddings, but the caption quality is still in an early stage.